

TASK A1:

[illegible]

TASK A2: (couldn't run : "ps -fp 1745" as it's a linux command and i'm working on windows so i tried to look for windows substitutes)

```
C:\Users\ZOUNY>docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS        NAMES
0bf3d59eab6   ubuntu:22.04 "bash"              10 minutes ago Up 10 minutes          nsdemo
870c16d4ffb1   ubuntu   "/bin/bash"         4 hours ago   Up 3 hours          test-container

C:\Users\ZOUNY>docker inspect nsdemo --format "{{.State.Pid}}"
1745

C:\Users\ZOUNY>ps -fp <1745>
The syntax of the command is incorrect.

C:\Users\ZOUNY>docker top nsdemo

```

UID	PID	PPID	C	STIME	T
root	1745	1725	0	16:15	?

```
C:\Users\ZOUNY>Get-Process -Id <1745>
The syntax of the command is incorrect.

C:\Users\ZOUNY>C:\Users\ZOUNY>Get-Process -Id <1745>
The syntax of the command is incorrect.

C:\Users\ZOUNY>The syntax of the command is incorrect
'The' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\ZOUNY>Get-Process -Id 1745
'Get-Process' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\ZOUNY>Get-Process -Id 1745
'Get-Process' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\ZOUNY>tasklist /FI "PID eq 1745"
INFO: No tasks are running which match the specified criteria.
```

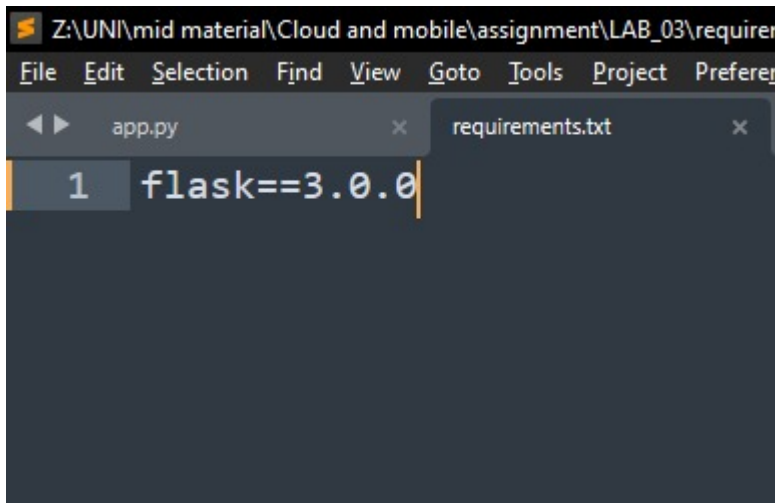
On the host: The exact same process gets a standard, unique PID.

TASK A3: (Starting a CPU- and memory-limited container and installing stress and generated load)

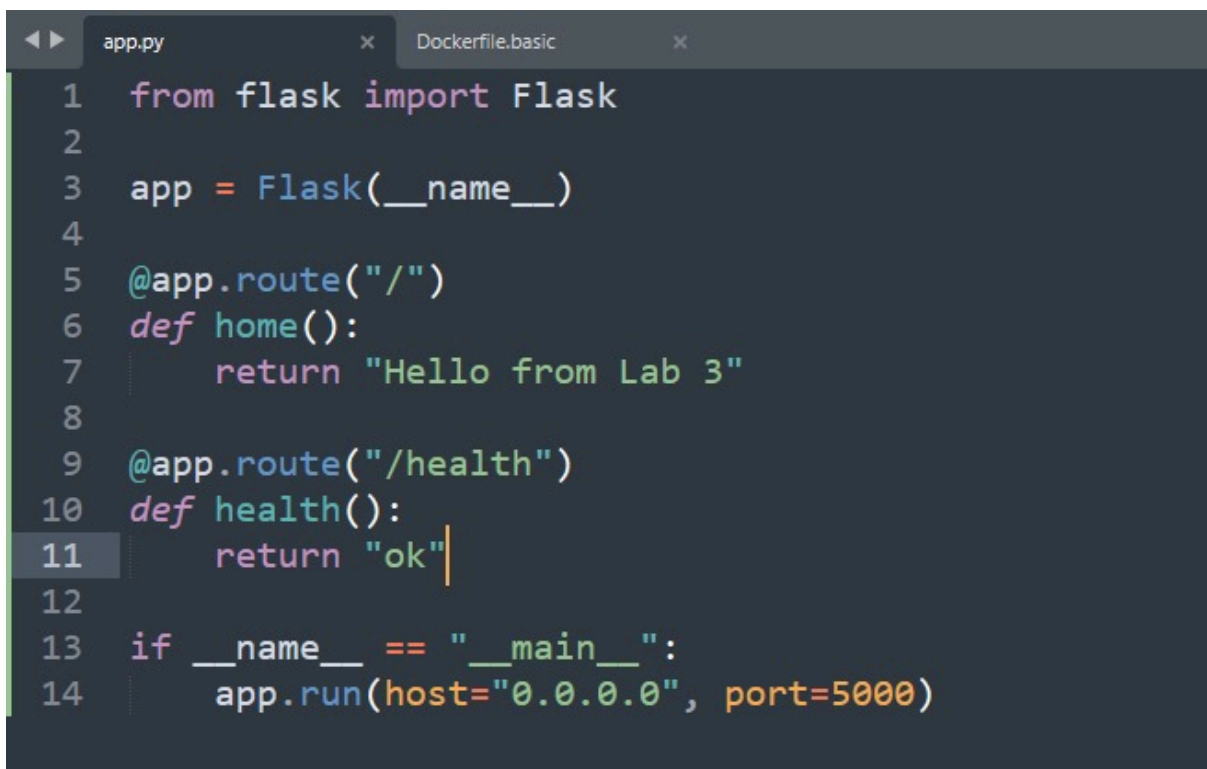
```
C:\Users\ZOUNY>docker run --rm -it --cpus="0.5" --memory="256m" ubuntu:22.04 bash
root@8411f7fb9195:/# apt-get update && apt-get install -y stress
Get:1 http://archive.ubuntu.com/ubuntu jammy InRelease [270 kB]
Get:2 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:4 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [6862 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy/multiverse amd64 Packages [266 kB]
Get:7 http://archive.ubuntu.com/ubuntu jammy/restricted amd64 Packages [164 kB]
Get:8 http://archive.ubuntu.com/ubuntu jammy/main amd64 Packages [1792 kB]
Get:9 http://archive.ubuntu.com/ubuntu jammy/universe amd64 Packages [17.5 MB]
Get:10 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [3863 kB]
Get:11 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 Packages [61.6 kB]
Get:12 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [1292 kB]
Get:13 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [4191 kB]
Get:14 http://archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 Packages [69.9 kB]
Get:15 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1601 kB]
Get:16 http://archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [7066 kB]
Get:17 http://archive.ubuntu.com/ubuntu jammy-backports/main amd64 Packages [82.7 kB]
Get:18 http://archive.ubuntu.com/ubuntu jammy-backports/universe amd64 Packages [35.7 kB]
Fetched 45.5 MB in 23s (1939 kB/s)
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  stress
0 upgraded, 1 newly installed, 0 to remove and 1 not upgraded.
Need to get 18.4 kB of archives.
After this operation, 52.2 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy/universe amd64 stress amd64 1.0.5-1 [18.4 kB]
Fetched 18.4 kB in 0s (86.0 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package stress.
(Reading database ... 4393 files and directories currently installed.)
Preparing to unpack .../stress_1.0.5-1_amd64.deb ...
Unpacking stress (1.0.5-1) ...
Setting up stress (1.0.5-1) ...
root@8411f7fb9195:/# stress --cpu 2 --vm 1 --vm-bytes 220M --timeout 30
stress: info: [235] dispatching hogs: 2 cpu, 0 io, 1 vm, 0 hdd
stress: info: [235] successful run completed in 30s
root@8411f7fb9195:/#
```

```
C:\Users\ZOUNY>docker stats
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT   MEM %     NET I/O     BLOCK I/O  PIDS
8411f7fb9195   cranky_bohr   0.04%     75.78MiB / 256MiB   29.60%    47.6MB / 992kB   0B / 0B    1
0bff3d59eab6   nsdemo        0.05%     4.344MiB / 7.635GiB 0.06%     49.1MB / 1.01MB  0B / 0B    1
870c16d4ffb1   test-container 0.00%     3.031MiB / 7.635GiB 0.04%     1.13kB / 0B      0B / 0B    1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT   MEM %     NET I/O     BLOCK I/O  PIDS
8411f7fb9195   cranky_bohr   0.00%     75.78MiB / 256MiB   29.60%    47.6MB / 992kB   0B / 0B    1
0bff3d59eab6   nsdemo        0.00%     4.344MiB / 7.635GiB 0.06%     49.1MB / 1.01MB  0B / 0B    1
870c16d4ffb1   test-container 0.00%     3.031MiB / 7.635GiB 0.04%     1.13kB / 0B      0B / 0B    1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT   MEM %     NET I/O     BLOCK I/O  PIDS
8411f7fb9195   cranky_bohr   0.00%     75.78MiB / 256MiB   29.60%    47.6MB / 992kB   0B / 0B    1
0bff3d59eab6   nsdemo        0.00%     4.344MiB / 7.635GiB 0.06%     49.1MB / 1.01MB  0B / 0B    1
870c16d4ffb1   test-container 0.00%     3.031MiB / 7.635GiB 0.04%     1.13kB / 0B      0B / 0B    1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT   MEM %     NET I/O     BLOCK I/O  PIDS
8411f7fb9195   cranky_bohr   0.00%     75.78MiB / 256MiB   29.60%    47.6MB / 992kB   0B / 0B    1
0bff3d59eab6   nsdemo        0.00%     4.344MiB / 7.635GiB 0.06%     49.1MB / 1.01MB  0B / 0B    1
870c16d4ffb1   test-container 0.00%     3.031MiB / 7.635GiB 0.04%     1.13kB / 0B      0B / 0B    1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT   MEM %     NET I/O     BLOCK I/O  PIDS
8411f7fb9195   cranky_bohr   0.00%     75.78MiB / 256MiB   29.60%    47.6MB / 992kB   0B / 0B    1
0bff3d59eab6   nsdemo        0.00%     4.344MiB / 7.635GiB 0.06%     49.1MB / 1.01MB  0B / 0B    1
870c16d4ffb1   test-container 0.00%     3.031MiB / 7.635GiB 0.04%     1.13kB / 0B      0B / 0B    1
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT   MEM %     NET I/O     BLOCK I/O  PIDS
8411f7fb9195   cranky_bohr   0.00%     75.78MiB / 256MiB   29.60%    47.6MB / 992kB   0B / 0B    1
```

TASK B1: (creating [app.py](#) and requirements.txt folders {all folders are uploaded in the repo})



A screenshot of a code editor window. The title bar shows the file path: Z:\UNI\mid material\Cloud and mobile\assignment\LAB_03\requirements. The editor has two tabs: 'app.py' and 'requirements.txt'. The 'requirements.txt' tab is active, and it contains a single line of text: '1 flask==3.0.0'. The line number '1' is highlighted on the left margin.



A screenshot of a code editor window. The title bar shows the file path: Z:\UNI\mid material\Cloud and mobile\assignment\LAB_03\requirements. The editor has two tabs: 'app.py' and 'Dockerfile.basic'. The 'app.py' tab is active, and it contains the following Python code:

```
1  from flask import Flask
2
3  app = Flask(__name__)
4
5  @app.route("/")
6  def home():
7      return "Hello from Lab 3"
8
9  @app.route("/health")
10 def health():
11     return "ok"
12
13 if __name__ == "__main__":
14     app.run(host="0.0.0.0", port=5000)
```


TASK B2: (Create Dockerfile.basic)

```
Z:\UNI\mid material\Cloud and mobile\assignment\LAB_03\Dockerfile.basic - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help
app.py requirements.txt Dockerfile.basic
1 FROM python:3.12-slim
2 WORKDIR /app
3 COPY requirements.txt .
4 RUN pip install -r requirements.txt
5 COPY app.py .
6 EXPOSE 5000
7 CMD ["python", "app.py"]
```

(Build and inspect the image)

```
Z:\UNI\mid material\Cloud and mobile\assignment\LAB_03>docker build -f Dockerfile.basic -t lab3-basic .
[+] Building 19.6s (10/10) FINISHED
=> [internal] load build definition from Dockerfile.basic
=> => transferring dockerfile: 196B
=> [internal] load metadata for docker.io/library/python:3.12-slim
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/python:3.12-slim@sha256:46cb7cc2877e60fbd5e21a9ae6115c30ace7a077b9f8772da879e45
=> => resolve docker.io/library/python:3.12-slim@sha256:46cb7cc2877e60fbd5e21a9ae6115c30ace7a077b9f8772da879e45
=> => sha256:cb8c0a9140ac107575972113eb5fd2e6c3cdddc592371a5ac31c4623dcb13d5d 1.29MB / 1.29MB
=> => sha256:3fc9d9ab5045181fbd49d1d30a2dbad144172d4c3c141f70117ddb0b918d8357 248B / 248B
=> => sha256:4d873bcef452ab1c054b9a0f25ace04a446c558a7b925aed2fc5bc21533e0f65 12.10MB / 12.10MB
=> => extracting sha256:cb8c0a9140ac107575972113eb5fd2e6c3cdddc592371a5ac31c4623dcb13d5d
=> => extracting sha256:4d873bcef452ab1c054b9a0f25ace04a446c558a7b925aed2fc5bc21533e0f65
=> => extracting sha256:3fc9d9ab5045181fbd49d1d30a2dbad144172d4c3c141f70117ddb0b918d8357
=> [internal] load build context
=> => transferring context: 310B
=> [2/5] WORKDIR /app
=> [3/5] COPY requirements.txt .
=> [4/5] RUN pip install -r requirements.txt
=> [5/5] COPY app.py .
=> => exporting to image
=> => exporting layers
=> => exporting manifest sha256:6c30da48e31ff073f874ff7b6c110dca597afc638e1422a257ef943ad4e86bba
=> => exporting config sha256:519ba2ce393f2faa3562e933e88f310c0766f156964fb22768e8c89eed06ae8
=> => exporting attestation manifest sha256:cd6882193f0699a4e0d03d883774106181b7fea01acca23c2ae9c68364c5cc2a
=> => exporting manifest list sha256:b4bce80bcf4fd5a928ba7f822c6bc4aebf5d824d63edb1179d8bfff1b0b8446a
=> => naming to docker.io/library/lab3-basic:latest
=> => unpacking to docker.io/library/lab3-basic:latest

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/imvejvp03jkj0pbz3nznay1w

Z:\UNI\mid material\Cloud and mobile\assignment\LAB_03>docker image ls
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
lab3-basic           latest          bb4ce80bcf4f   2 minutes ago   190MB
redis               latest          832d7785830f   2 days ago      202MB
ubuntu              22.04           962f6cadeae0   2 weeks ago     117MB
ubuntu              latest          c4a8d5503dfb   2 weeks ago     117MB
hello-world         latest          f9078146db2e   4 weeks ago     20.4kB
adventureworksvs-api latest          32d8d60bf15b   16 months ago   406MB
scm2-rest           latest          a9c949f96f73   16 months ago   354MB
scm2-grpc           latest          681842bcee8     16 months ago   330MB
scm2-corewcmigration latest          41b778a97420   16 months ago   340MB
educational-management-system-e-assistant-api latest          4eed602c6154   17 months ago   302MB
e-assistant         latest          f078639398c4   17 months ago   302MB
folders-server2     latest          51e43ff76603   17 months ago   208MB
my-flask-app        latest          42ffe3c53744   17 months ago   208MB
folders-server1     latest          469b61ad0074   17 months ago   208MB
grafana/grafana     latest          fa801ab6e1ae   17 months ago   646MB
prom/prometheus     latest          3b9b2a15d376   17 months ago   410MB
mcr.microsoft.com/mssql/server latest          45a1a9d13ca5   17 months ago   2.19GB
docker/welcome-to-docker latest          eedaff45e3c7   2 years ago     29.5MB
gcr.io/etcd-development/etcd v3.5.0         28759af54acd   4 years ago     156MB

Z:\UNI\mid material\Cloud and mobile\assignment\LAB_03>docker history lab3-basic
IMAGE             CREATED          CREATED BY          SIZE      COMMENT
bb4ce80bcf4f      2 minutes ago   CMD ["python" "app.py"] 0B        buildkit.dockerfile.v0
<missing>         2 minutes ago   EXPOSE map[5000/tcp:{}] 0B        buildkit.dockerfile.v0
<missing>         2 minutes ago   COPY app.py . # buildkit 12.3kB    buildkit.dockerfile.v0
<missing>         2 minutes ago   RUN /bin/sh -c pip install -r requirements.txt 9.7MB     buildkit.dockerfile.v0
<missing>         2 minutes ago   COPY requirements.txt . # buildkit 8.19kB    buildkit.dockerfile.v0
<missing>         2 minutes ago   WORKDIR /app         8.19kB    buildkit.dockerfile.v0
<missing>         2 days ago      CMD ["python3"]        0B        buildkit.dockerfile.v0
```


TASK B3: (Create Dockerfile.multistage)

```
Z:\UNI\mid material\Cloud and mobile\assignment\LAB_03\Dockerfile.multistage - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help

app.py requirements.txt Dockerfile.basic Dockerfile.multistage

1 FROM python:3.12-slim AS builder
2 WORKDIR /build
3 COPY requirements.txt .
4 RUN pip install --prefix=/install -r requirements.txt
5
6 FROM python:3.12-slim
7 WORKDIR /app
8 COPY --from=builder /install /usr/local
9 COPY app.py .
10 EXPOSE 5000
11 CMD ["python", "app.py"]
```

(Build and compare)

```
[+] Building 16.8s (12/12) FINISHED
=> [internal] load build definition from Dockerfile.multistage
=> => transferring dockerfile: 312B
=> [internal] load metadata for docker.io/library/python:3.12-slim
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 85B
=> CACHED [builder 1/4] FROM docker.io/library/python:3.12-slim@sha256:46cb7cc2877e60fbd5e21a9ae6115c30ace7a077b9f8772da8
=> => resolve docker.io/library/python:3.12-slim@sha256:46cb7cc2877e60fbd5e21a9ae6115c30ace7a077b9f8772da879e4590c18c2e3
=> CACHED [stage-1 2/4] WORKDIR /app
=> [builder 2/4] WORKDIR /build
=> [builder 3/4] COPY requirements.txt .
=> [builder 4/4] RUN pip install --prefix=/install -r requirements.txt
=> [stage-1 3/4] COPY --from=builder /install /usr/local
=> [stage-1 4/4] COPY app.py .
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:e169717190ea1e33a1d4566e70123b11f22117029bd78656f50cc9c4c4a9cc8b
=> => exporting config sha256:4d78796945ed02ec476cf1f81b5064919b6199a14ce606348f88dee167305e2f
=> => exporting attestation manifest sha256:4da650616c80d9278b33f60ac1e34176df335db6cfffcc052aac8caf6152db4f3
=> => exporting manifest list sha256:e14e8e447c8dc8186ead180c8c273fc7b52dd1d2882025fb457c205df76c832
=> => naming to docker.io/library/lab3-multi:latest
=> => unpacking to docker.io/library/lab3-multi:latest
=> => # Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
View build details: docker-desktop://dashboard/build/linux/desktop-linux/x081eafy9itjuwbsxcp8fm8i2
=> => # Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Z:\UNI\mid material\Cloud and mobile\assignment\LAB_03>docker image ls

REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
lab3-multi           latest             e14e8e447c8d       About a minute ago 184MB
lab3-basic           latest             bb4ce80bcf4f       18 minutes ago    190MB
redis                latest             832d7785830f       2 days ago        202MB
ubuntu               22.04             962f6cadeae0       2 weeks ago       117MB
ubuntu               latest             c4a8d5503dfb       2 weeks ago       117MB
hello-world          latest             f9078146db2e       4 weeks ago       20.4kB
adventureworksvs-api latest             32d8d60bf15b       16 months ago     406MB
scm2-rest            latest             a9c949f96f73       16 months ago     354MB
scm2-grpc            latest             681842bceee8       16 months ago     330MB
scm2-coreworkmigration latest            41b778a97420       16 months ago     340MB
educational-management-system-e-assistant--api latest            4eed602c6154       17 months ago     302MB
e_assistant          latest             f078639398c4       17 months ago     302MB
folders-server2      latest             51e43ff76603       17 months ago     208MB
my-flask-app          latest             42ffe3c53744       17 months ago     208MB
folders-server1      latest             469b61ad0074       17 months ago     208MB
grafana/grafana      latest             fa801ab6e1ae       17 months ago     646MB
prom/prometheus      latest             3b9b2a15d376       17 months ago     410MB
mcr.microsoft.com/mssql/server latest            45a1a9d13ca5       17 months ago     2.19GB
docker/welcome-to-docker latest            eedaff45e3c7       2 years ago       29.5MB
gcr.io/etcd-development/etcd v3.5.0           28759af54acd       4 years ago       156MB

Z:\UNI\mid material\Cloud and mobile\assignment\LAB_03>docker history lab3-multi
IMAGE                CREATED              CREATED BY          SIZE      COMMENT
e14e8e447c8d        About a minute ago  CMD ["python" "app.py"]  0B        buildkit.dockerfile.v0
<missing>           About a minute ago  EXPOSE map[5000/tcp:{}]  0B        buildkit.dockerfile.v0
<missing>           About a minute ago  COPY app.py . # buildkit 12.3kB    buildkit.dockerfile.v0
<missing>           About a minute ago  COPY /install /usr/local # buildkit 5.51MB    buildkit.dockerfile.v0
<missing>           19 minutes ago     WORKDIR /app           8.19kB    buildkit.dockerfile.v0
<missing>           2 days ago         CMD ["python3"]        0B        buildkit.dockerfile.v0
<missing>           2 days ago         RUN /bin/sh -c set -eux; for src in idle3 p... 16.4kB    buildkit.dockerfile.v0
<missing>           2 days ago         RUN /bin/sh -c set -eux; savedAptMark="$({a... 41.3MB    buildkit.dockerfile.v0
<missing>           2 days ago         ENV PYTHON_SHA256=c08bc65a81971c1dd578318282... 0B        buildkit.dockerfile.v0
<missing>           2 days ago         ENV PYTHON_VERSION=3.12.13 0B        buildkit.dockerfile.v0
<missing>           2 days ago         ENV GPG_KEY=7169605F62C751356D054A26A821E680... 0B        buildkit.dockerfile.v0
<missing>           2 days ago         RUN /bin/sh -c set -eux; apt-get update; a... 4.94MB    buildkit.dockerfile.v0
<missing>           2 days ago         ENV LANG=C.UTF-8       0B        buildkit.dockerfile.v0
<missing>           2 days ago         ENV PATH=/usr/local/bin:/usr/local/sbin:/usr... 0B        buildkit.dockerfile.v0
<missing>           3 days ago         # debian.sh --arch 'amd64' out/ 'trixie' '@1... 87.4MB    debuerreotype 0.17
```

Questions: Which image is smaller? Which steps are cached if only app.py changes? Why does layer order matter?

Answer:

Smaller image: lab3-multi (184MB vs 190MB). It discards temporary build tools and cache files.

Cached steps: Everything before COPY app.py . stays cached (including the heavy dependency installation).

Layer order: Docker caches top-to-bottom. Changing a layer breaks the cache for all layers below it. Put rarely changed files at the top and frequently changed code at the bottom to speed up builds.

TASK C1: created a cluster and verified it's working and got the clusters info (switched to powershell cause it's faster)

```
PS C:\Users\ZOUNY> kind create cluster --name lab3-cluster --image kindest/node:v1.30.0
Creating cluster "lab3-cluster" ...
 * Ensuring node image (kindest/node:v1.30.0) @ ...
 * Ensuring node image (kindest/node:v1.30.0) @ ...
 * Preparing nodes @ ...
 * Preparing nodes @ ...
 * Writing configuration @ ...
 * Writing configuration @ ...
 * Starting control-plane @ ...
 * Starting control-plane @ ...
 * Installing CNI @ ...
 * Installing CNI @ ...
 * Installing StorageClass @ ...
 * Installing StorageClass @ ...
Set kubectl context to "kind-lab3-cluster"
You can now use your cluster with:

kubectl cluster-info --context kind-lab3-cluster

Not sure what to do next? @ Check out https://kind.sigs.k8s.io/docs/user/quick-start/
PS C:\Users\ZOUNY> kubectl cluster-info
Kubernetes is running at https://127.0.0.1:57835
CoreDNS is running at https://127.0.0.1:57835/api/v1/namespaces/kube-system/services/kube-dns:proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
PS C:\Users\ZOUNY> kubectl get nodes -o wide
NAME                                STATUS    ROLES    AGE   VERSION   INTERNAL-IP   EXTERNAL-IP   OS-IMAGE                                     KERNEL-VERSION   CONTAINER-RUNTIME
lab3-cluster-control-plane          Ready    control-plane   38s   v1.30.0   172.18.0.2    <none>         Debian GNU/Linux 12 (bookworm)            5.15.167.4-microsoft-standard-WSL2   containerd://1.7.15
PS C:\Users\ZOUNY>
```

TASK C2: (loaded the basic and multi-stage images into kind)

```
PS C:\Users\ZOUNY> kind load docker-image lab3-basic --name lab3-cluster
Image: "lab3-basic" with ID "sha256:bb4ce80bcf4f6d5a928ba7f822c6bc4aebf5d824d63edb1179d8bffb0b8446a" not yet present on node "lab3-cluster-control-plane", loading...
PS C:\Users\ZOUNY> kind load docker-image lab3-multi --name lab3-cluster
ERROR: no nodes found for cluster "lab3-cluster\"
PS C:\Users\ZOUNY> kind load docker-image lab3-multi --name lab3-cluster
Image: "lab3-multi" with ID "sha256:e14e8e447c8dc8186eaad180c8c273fc7b52d1d12882025fb457c205df76c832" not yet present on node "lab3-cluster-control-plane", loading...
PS C:\Users\ZOUNY>
```

TASK C3: (created deployment.yaml)

```
app.py x requirements.txt x Dockerfile.basic x Dockerfile.multistage x deployment.yaml x
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: lab3-web
5  spec:
6    replicas: 3
7    selector:
8      matchLabels:
9        app: lab3-web
10   template:
11     metadata:
12       labels:
13         app: lab3-web
14     spec:
15       containers:
16       - name: web
17         image: lab3-multi
18         imagePullPolicy: Never
19         ports:
20         - containerPort: 5000
```

(created service.yaml)

```
app.py x requirements.txt x Dockerfile.basic x Dockerfile.multistage x deployment.yaml x service.yaml x
1  apiVersion: v1
2  kind: Service
3  metadata:
4    name: lab3-web-svc
5  spec:
6    selector:
7      app: lab3-web
8    ports:
9    - port: 80
10      targetPort: 5000
11    type: ClusterIP
```

(deploy)

```
Z:\UNI\mid material\Cloud and mobile\assignment\LAB_03>kubectl apply -f deployment.yaml
deployment.apps/lab3-web created

Z:\UNI\mid material\Cloud and mobile\assignment\LAB_03>kubectl apply -f service.yaml
service/lab3-web-svc created
```


(inspect)

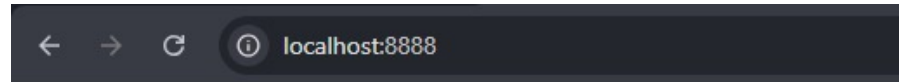
```
Z:\UNI\mid material\Cloud and mobile\assignment\LAB_03>kubect1 get pods -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP            NODE                                NOMINATED NODE   READINESS GATES
lab3-web-5d56ff7f5b-rzv5r           1/1     Running   0           2m57s  10.244.0.6    lab3-cluster-control-plane         <none>            <none>
lab3-web-5d56ff7f5b-vzjq9           1/1     Running   0           2m57s  10.244.0.7    lab3-cluster-control-plane         <none>            <none>
lab3-web-5d56ff7f5b-zgr8j           1/1     Running   0           2m57s  10.244.0.5    lab3-cluster-control-plane         <none>            <none>

Z:\UNI\mid material\Cloud and mobile\assignment\LAB_03>kubect1 get deployments
NAME      READY   UP-TO-DATE   AVAILABLE   AGE
lab3-web  3/3     3             3            3m3s

Z:\UNI\mid material\Cloud and mobile\assignment\LAB_03>kubect1 describe deployment lab3-web
Name:                lab3-web
Namespace:           default
CreationTimestamp:    Fri, 24 Apr 2026 21:28:59 +0300
Labels:              <none>
Annotations:         deployment.kubernetes.io/revision: 1
Selector:             app=lab3-web
Replicas:            3 desired | 3 updated | 3 total | 3 available | 0 unavailable
StrategyType:        RollingUpdate
MinReadySeconds:     0
RollingUpdateStrategy: 25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=lab3-web
  Containers:
    web:
      Image:      lab3-multi
      Port:       5000/TCP
      Host Port:  0/TCP
      Environment: <none>
      Mounts:      <none>
  Volumes:      <none>
  Node-Selectors: <none>
  Tolerations:  <none>
Conditions:
  Type           Status  Reason
  ----           -
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
OldReplicaSets: <none>
NewReplicaSet:  lab3-web-5d56ff7f5b (3/3 replicas created)
Events:
  Type           Reason             Age    From              Message
  ----           -
  Normal        ScalingReplicaSet  3m9s   deployment-controller  Scaled up replica set lab3-web-5d56ff7f5b to 3
```

TASK C4:(Accessing the service)

```
Z:\UNI\mid material\Cloud and mobile\assignment\LAB_03>kubectl port-forward service/lab3-web-svc 8888:80
Forwarding from 127.0.0.1:8888 -> 5000
Forwarding from [::1]:8888 -> 5000
Handling connection for 8888
Handling connection for 8888
Handling connection for 8888
Handling connection for 8888
```



Hello from Lab 3

```
PS C:\Users\ZOUNY> curl.exe http://localhost:8888/
Hello from Lab 3
PS C:\Users\ZOUNY> curl.exe http://localhost:8888/health
ok
PS C:\Users\ZOUNY>
```

TASK D1: (Labeling the node)

```
Z:\UNI\mid material\Cloud and mobile\assignment\LAB_03>kubectl label nodes --all node-role=general
node/lab3-cluster-control-plane labeled

Z:\UNI\mid material\Cloud and mobile\assignment\LAB_03>kubectl get nodes --show-labels
NAME                                STATUS    ROLES    AGE   VERSION   LABELS
lab3-cluster-control-plane          Ready    control-plane   32m   v1.30.0   beta.kubernetes.io/arch=amd64,beta.kubernetes.io/os=linux,kubernetes.io/arch=amd64,kubernetes.io/hostname=lab3-cluster-control-plane,kubernetes.io/os=linux,node-role.kubernetes.io/control-plane=,node-role=general
```

TASK D2: (edited deployment.yaml)

```
1 apiVersion: apps/v1
2 kind: Deployment
3 metadata:
4   name: lab3-web
5 spec:
6   replicas: 3
7   selector:
8     matchLabels:
9       app: lab3-web
10  template:
11    metadata:
12      labels:
13        app: lab3-web
14    spec:
15      containers:
16      - name: web
17        image: lab3-multi
18        imagePullPolicy: Never
19        ports:
20        - containerPort: 5000
21      nodeSelector:
22        node-role: general
```

(Re-applying the Deployment and inspect placement again)

```
Z:\UNI\mid material\Cloud and mobile\assignment\LAB_03>kubectl apply -f deployment.yaml
deployment.apps/lab3-web configured

Z:\UNI\mid material\Cloud and mobile\assignment\LAB_03>kubectl get pods -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP              NODE                                NOMINATED NODE   READINESS GATES
lab3-web-5d56ff7f5b-rzv5r           1/1     Terminating    0        38m   10.244.0.6      lab3-cluster-control-plane         <none>            <none>
lab3-web-5d56ff7f5b-vzjq9           1/1     Terminating    0        38m   10.244.0.7      lab3-cluster-control-plane         <none>            <none>
lab3-web-5d56ff7f5b-zgr8j           1/1     Terminating    0        38m   10.244.0.5      lab3-cluster-control-plane         <none>            <none>
lab3-web-c457fd886-dggk8            1/1     Running         0         4s    10.244.0.9      lab3-cluster-control-plane         <none>            <none>
lab3-web-c457fd886-dxqr5            1/1     Running         0         3s    10.244.0.10     lab3-cluster-control-plane         <none>            <none>
lab3-web-c457fd886-j7cv5            1/1     Running         0         4s    10.244.0.8      lab3-cluster-control-plane         <none>            <none>
```

Question: Why is this different from hard-coding a machine name in traditional deployment models? How does declarative scheduling improve resilience and maintainability?

Answer:

No Hard-coding: You are targeting machines by *labels* (traits/roles) instead of specific hostnames. You ask for a "type" of machine, not a specific one.

Resilience: If a server crashes, Kubernetes automatically finds another healthy node with the same label and moves the workload there.

Maintainability: You can add, remove, or replace physical servers in your cluster at any time without ever needing to update your deployment code.

TASK E1: (Triggering self-healing by deleting a Pod)

```
Z:\UNI\mid material\Cloud and mobile\assignment\LAB_03>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
lab3-web-9d9b56bd8-bfj8v           1/1     Running   0           8s
lab3-web-9d9b56bd8-vzwnv           1/1     Running   0          10s
lab3-web-9d9b56bd8-xgfh5           1/1     Running   0           7s
lab3-web-c457fd886-dggk8           1/1     Terminating 0         6m57s
lab3-web-c457fd886-dxqr5           1/1     Terminating 0         6m56s
lab3-web-c457fd886-j7cv5           1/1     Terminating 0         6m57s

Z:\UNI\mid material\Cloud and mobile\assignment\LAB_03>kubectl delete pod lab3-web-9d9b56bd8-bfj8v
pod "lab3-web-9d9b56bd8-bfj8v" deleted

Z:\UNI\mid material\Cloud and mobile\assignment\LAB_03>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
lab3-web-9d9b56bd8-fxgkn           1/1     Running   0          49s
lab3-web-9d9b56bd8-vzwnv           1/1     Running   0          103s
lab3-web-9d9b56bd8-xgfh5           1/1     Running   0          100s
```

Question: Why is this called reconciliation rather than simple restart logic?

Answer:

Restart logic: Reacts to a crash by restarting the same process.

Reconciliation: Constantly compares your desired state to the current state. When you delete a pod, it creates a brand new one simply to make the numbers match again.

TASK E2: (created probe-deployment.yaml)

```
probe-deployment.yaml x
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: lab3-web-probe
5  spec:
6    replicas: 2
7    selector:
8      matchLabels:
9        app: lab3-web-probe
10   template:
11     metadata:
12       labels:
13         app: lab3-web-probe
14     spec:
15       containers:
16       - name: web
17         image: lab3-multi
18         imagePullPolicy: Never
19         ports:
20         - containerPort: 5000
21         readinessProbe:
22           httpGet:
23             path: /health
24             port: 5000
25           initialDelaySeconds: 3
26           periodSeconds: 5
27         livenessProbe:
28           httpGet:
29             path: /health
30             port: 5000
31           initialDelaySeconds: 10
32           periodSeconds: 10
```

(added readiness and liveness probes)

```
Z:\UNI\mid material\Cloud and mobile\assignment\LAB_03>kubectl apply -f probe-deployment.yaml
deployment.apps/lab3-web-probe created

Z:\UNI\mid material\Cloud and mobile\assignment\LAB_03>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
lab3-web-9d9b56bd8-fxgkn            1/1     Running   0           6m58s
lab3-web-9d9b56bd8-vzwnv            1/1     Running   0           7m52s
lab3-web-9d9b56bd8-xgfh5            1/1     Running   0           7m49s
lab3-web-probe-5db74dcbc7-nzlhq     1/1     Running   0           9s
lab3-web-probe-5db74dcbc7-qmkz2     1/1     Running   0           9s

Z:\UNI\mid material\Cloud and mobile\assignment\LAB_03>

Z:\UNI\mid material\Cloud and mobile\assignment\LAB_03>kubectl describe pod lab3-web-9d9b56bd8-fxgkn
Name:                                lab3-web-9d9b56bd8-fxgkn
Namespace:                           default
Priority:                              0
Service Account:                      default
Node:                                 lab3-cluster-control-plane/172.18.0.2
Start Time:                           Fri, 24 Apr 2026 22:15:18 +0300
Labels:                               app=lab3-web
                                      pod-template-hash=9d9b56bd8
Annotations:                           <none>
Status:                               Running
IP:                                   10.244.0.14
IPs:                                  10.244.0.14
Controlled By:                        ReplicaSet/lab3-web-9d9b56bd8
Containers:
  web:
    Container ID:   containerd://7b0b2c95c5b16ad71517154d8bf425488d10f92cdc01550f519d3fb4e8faa176
    Image:          lab3-multi
    Image ID:       docker.io/library/import-2026-04-24@sha256:e14e8e447c8dc8186eaad180c8c273fc7b52d1d12882025fb457c205df76c832
    Port:           5000/TCP
    Host Port:      0/TCP
    State:          Running
      Started:      Fri, 24 Apr 2026 22:15:18 +0300
    Ready:          True
    Restart Count:  0
    Liveness:       http-get http://:5000/health delay=5s timeout=1s period=5s #success=1 #failure=3
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-hs96p (ro)
Conditions:
  Type              Status
  PodReadyToStartContainers  True
  Initialized        True
  Ready              True
  ContainersReady    True
  PodScheduled       True
Volumes:
  kube-api-access-hs96p:
    Type:              Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName:       kube-root-ca.crt
    ConfigMapOptional:   <nil>
    DownwardAPI:        true
  QoS Class:           BestEffort
  Node-Selectors:      node-role=general
  Tolerations:         node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                      node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type    Reason      Age    From          Message
  ----    -
  Normal  Scheduled   7m26s  default-scheduler  Successfully assigned default/lab3-web-9d9b56bd8-fxgkn to lab3-cluster-control-plane
  Normal  Pulled      7m26s  kubelet        Container image "lab3-multi" already present on machine
  Normal  Created     7m26s  kubelet        Created container web
  Normal  Started     7m26s  kubelet        Started container web
```

TASK E3: (I used the command and tried various ways to get the results but i couldn't get the restart counter to go up)

```
PS C:\Users\ZOUNY> kubectl get pods -w
NAME                                READY   STATUS    RESTARTS   AGE
lab3-web-9d9b56bd8-fxgkn            1/1     Running   0           16m
lab3-web-9d9b56bd8-vzwnv            1/1     Running   0           16m
lab3-web-9d9b56bd8-xgfh5            1/1     Running   0           16m
lab3-web-probe-5db74dcbc7-nzlhq     1/1     Running   0           9m13s
lab3-web-probe-5db74dcbc7-qmkz2     1/1     Running   0           9m13s

Z:\UNI\mid material\Cloud and mobile\assignment\LAB_03>kubectl exec -it lab3-web-9d9b56bd8-fxgkn -- /bin/sh
# ps -ef
/bin/sh: 1: ps: not found
# kill 1
# exit
```

REFLECTION QUESTIONS:

1. Why do namespaces alone not guarantee fair resource use?

-Answer:Namespaces only isolate visibility (what a process can see), not resources (what a process can consume).

2. How do cgroups improve cluster stability?

-Answer:They enforce hard limits on CPU and memory usage, preventing a single rogue container from starving others or crashing the host

3. Why is Docker image layering important for large-scale orchestration?

-Answer:It saves storage space and drastically speeds up deployments by caching and reusing common base layers across multiple containers.

4. What does Kubernetes mean by desired state?

-Answer:The exact configuration you declare. Kubernetes continuously works in the background to make the actual state match this requested state.

5. How is self-healing different from traditional manual operations?

-Answer:It automatically detects failures and replaces broken components instantly without requiring any human intervention

6. Why are readiness probes and liveness probes not interchangeable?

-Answer:A liveness probe restarts an app if it is completely dead/stuck. A readiness probe temporarily stops sending network traffic to an app if it is just busy or loading.

7. What is one limitation of observing scheduling in a single-node kind cluster compared with a real multi node cluster?

-Answer:You cannot observe multi-node behaviors like node failover, node-affinity scheduling rules, or physical network latency.